



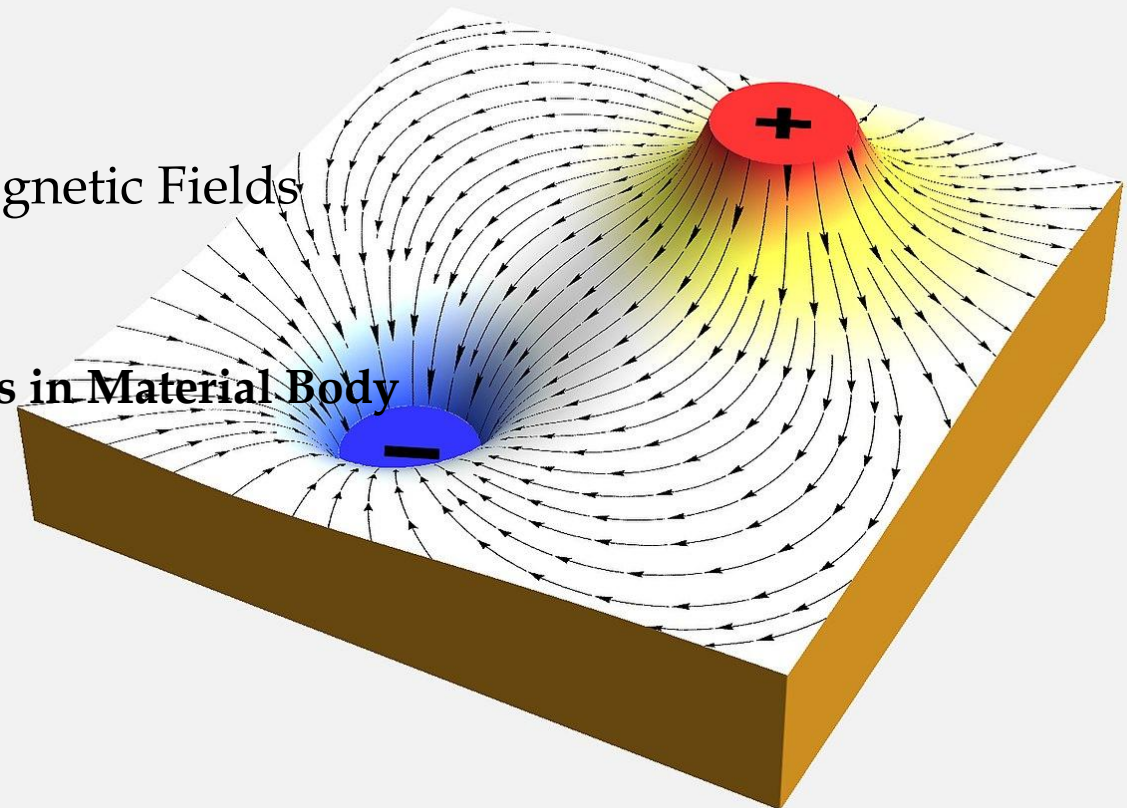
Addis Ababa University

Collage of Technology and Built Environment (CTBE)

School of Electrical and Computer Engineering

ECEG 2102: Electromagnetic Fields

Module 3: Electrostatic Fields in Material Body



Introduction and Objective

- **Bridge from Previous Modules:**
 - We've established the fundamental laws of electrostatics in **free space (the vacuum)**.
 - This allowed us to calculate electric fields produced by different charge distributions.
 - We have also dealt with potential and Energy considerations again in free space.
- **New Focus:**
 - Now, we introduce **material media** — specifically *conductors* and *dielectrics* — to understand how the **presence of matter modifies the electric field**.
- **Core Goal:**
 - To develop the equations and physical understanding required to analyze **electrostatic phenomena in real-world materials**.
- **Learning Outcomes:**
 - Differentiate between **convection** and **conduction** currents.
 - Describe **polarization** mechanisms in dielectrics.
 - Relate **$\mathbf{P}$** , **$\mathbf{E}$** , and **$\mathbf{D}$** through material properties.
 - Apply **boundary conditions** across dielectric and conductor interfaces.
 - Recognize real-world dielectric breakdown and insulation behavior.



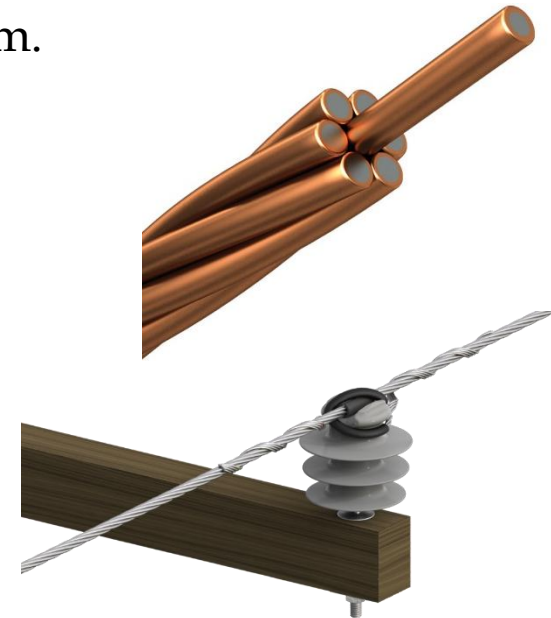
Why Study Electric Fields in Materials?

- Most electric fields in practice exist *within materials*, not in vacuum.
- Materials affect:
 - **Current flow** (conductivity)
 - **Energy storage** (permittivity)
 - **Field shape and continuity** (boundaries)
- Real-world examples:
 - Capacitors
 - Insulators
 - Cables
 - Sensors
- **Analogy:**
 - “The electric field is like the wind — materials are the terrain that shape its path.”



Electrical Classification of Materials

- Electric materials are classified according to how easily charges move through them.
- **Conductors (Metals)**
 - **Very high conductivity** ($\sigma \gg 1$), typically 10^6 – 10^8 S/m.
 - Contain a large number of **free electrons** not bound to specific atoms.
 - Charges can move freely under the influence of an electric field.
 - **Examples:** Copper, silver, aluminum.
 - **Engineering note:** Resistivity $\rho = 1/\sigma$ is very small – ideal for low-loss current flow.
- **Dielectrics (Insulators)**
 - **Very low conductivity** ($\sigma \ll 1$), typically 10^{-12} – 10^{-16} S/m.
 - Electrons are **tightly bound** to their atoms; very few are available for conduction.
 - They cannot support steady (DC) current, but store energy by **polarization**.
 - **Examples:** Glass, rubber, mica, quartz.
 - **Applications:** Cable insulation, capacitor dielectrics.
- **Semiconductors**
 - Conductivity lies **between conductors and dielectrics** (10^{-6} – 10^2 S/m).
 - Charge carrier density depends on **doping, temperature, and electric fields**.
 - **Examples:** Silicon, germanium.
 - **Applications:** Diodes, transistors, integrated circuits.





Electric Current

- The **current** (in Amperes) through a given area is the electric charge passing through the area per unit time.
- Thus, in a current of one ampere, charge is being transferred at a rate of one coulomb per second.

$$I = \frac{dQ}{dt}$$

- **Current Density J**: if current I flows through a planar surface ΔS , the current density is:

$$J = \frac{\Delta I}{\Delta S}$$

assuming that the current density is perpendicular to the surface. If the current density is not normal to the surface:

$$\Delta I = \vec{J} \cdot \vec{a}_n \Delta S = \vec{J} \cdot \Delta \vec{S}$$

- Thus, the total current flowing through a surface S is:

$$I = \int_S \vec{J} \cdot d\vec{S}$$

- Depending on how I is produced, there are different kinds of current density:
 - **convection current** density,
 - **conduction current** density, and
 - displacement current density.
- We will consider convection and conduction current densities here; displacement current density will be considered in future courses.
- The equation $I = \int_S \vec{J} \cdot d\vec{S}$ applies to any kind of current density.
- Compared with the general definition of flux $\Psi = \int_S \vec{D} \cdot d\vec{S}$ we can see that the current I through S is merely the flux of the current density $\vec{J}$.



Convection Current

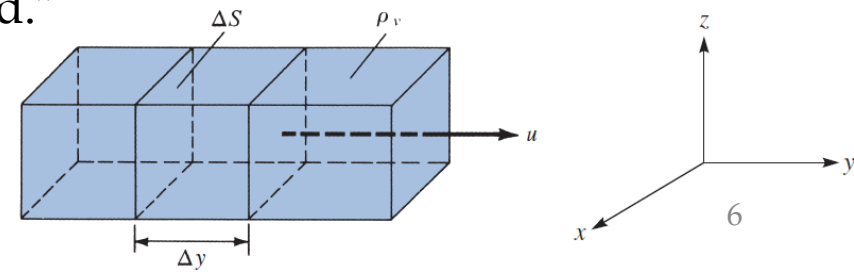
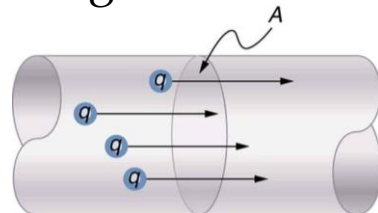
- Flow of **free charges** through an **insulating** medium (gas, vacuum, or liquid).
- Does not involve conductors and consequently, does not satisfy Ohm's law.

$$\Delta I = \frac{\Delta Q}{\Delta t} = \frac{\rho_v \Delta v}{\Delta t} = \frac{\rho_v \Delta S \Delta y}{\Delta t} = \rho_v \Delta S \frac{\Delta y}{\Delta t} = \rho_v \Delta S u$$

The current density: J

$$J = \frac{\Delta I}{\Delta S} = \rho_v u$$

- In general,
- Examples:
 - Electron beam in CRT
 - Ion flow in plasma
- **Analogy:**
 - “Like raindrops flying through air — moving freely, not confined.”



Conduction Current

- Caused by **drift motion** of electrons in a conductor under electric field.
- Conduction current **requires a conductor**.
- A conductor is characterized by a large number of free electrons that provide conduction current due to an impressed electric field.
- When an electric field $\vec{E}$ is applied, the force on an electron with charge $-e$ is

$$\vec{F} = -e\vec{E}$$
- If an electron with mass m is moving in an electric field $\vec{E}$ with an average drift velocity $\vec{u}$, according to Newton's law, the average change in momentum of the free electron must match the applied force. Thus:

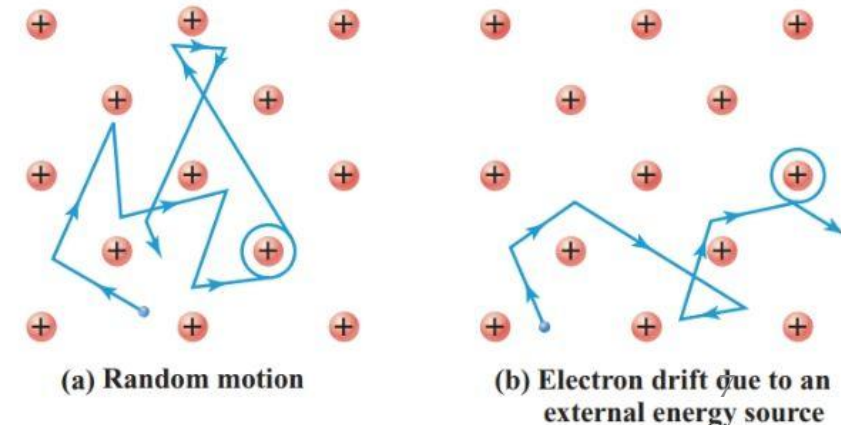
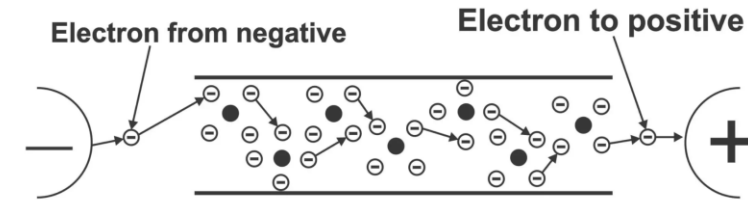
$$\frac{m\vec{u}}{\tau} = -e\vec{E} \Rightarrow \vec{u} = \frac{-e\tau}{m}\vec{E}$$

where τ is the average time interval between collisions. If there are n electrons per unit volume, the electronic charge density is given by $\rho_v = -ne$. Thus, the *conduction current density* is:

$$\vec{J} = \rho_v \vec{u} = \frac{ne^2\tau}{m}\vec{E} = \sigma\vec{E}$$

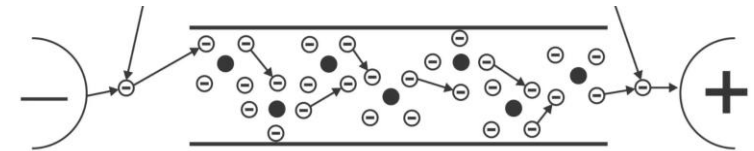
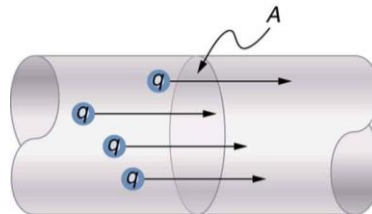
where $\sigma = \frac{ne^2\tau}{m}$ is the conductivity of the conductor which depends on material & temperature

- This is point form of **Ohm's Law**, which can be stated as: $\vec{J}$ and $\vec{E}$ are parallel, and the ratio $\frac{|\vec{J}|}{|\vec{E}|}$ is constant.
- Example:** copper wire in a circuit.



Convection vs Conduction

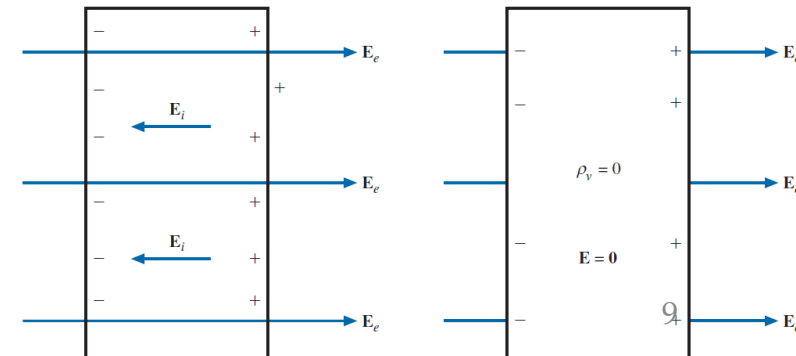
Property	Convection	Conduction
Medium	Insulator / gas / vacuum	Metal or semiconductor
Carrier motion	Free particles	Drift of electrons
Ohm's Law	✗ Not obeyed	✓ ($\vec{J} = \sigma \vec{E}$)
Example	Electron beam	Electric wire current



Perfect Conductors

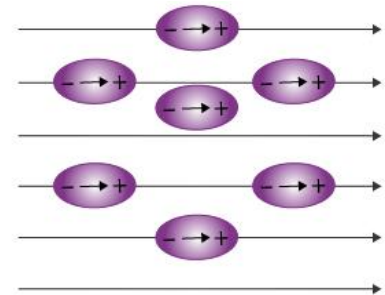
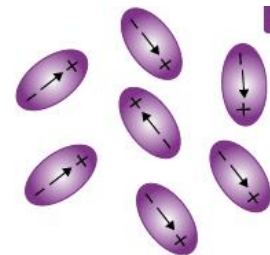
- A conductor has an abundance of charge that is free to move.
- When an external electric field $\vec{E}_e$ is applied, the positive free charges are pushed along the same direction as the applied field, while the negative free charges move in the opposite direction.
- This charge migration takes place very quickly. The free charges do two things:
 - First, they accumulate on the surface of the conductor and form an *induced surface charge*.
 - Second, the induced charges set up an internal induced field $\vec{E}_i$, which cancels the externally applied field $\vec{E}_e$.
- This leads to an important property of a conductor: The electrostatic field inside a **perfect conductor** ($\sigma = \infty$) is zero.
- As a result, a perfect conductor is called an *equipotential* body, implying that the potential is the same everywhere in the conductor. This is based on the fact that $\vec{E} = -\nabla V = 0$.
- In perfect conductor $\sigma = \infty$, considering Ohm's law, $\vec{J} = \sigma \vec{E}$, to maintain a finite current density $\vec{J}$, it requires that the electric field inside the conductor $\vec{E} \rightarrow 0$ vanish.
- If some charges are introduced in the interior of such a conductor, the charges will move to the surface and redistribute themselves quickly in such a manner that the field inside the conductor vanishes.
- According to Gauss's law, if $\vec{E} = 0$, the charge density ρ_v must be zero.
- In summary: for a perfect conductor:

$$\sigma = \infty, \vec{E} = 0, \rho_v = 0, V \text{ is constant}, V_{AB} = 0$$

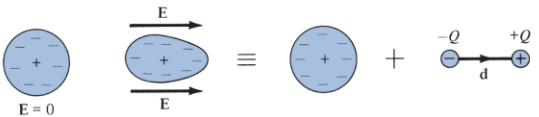
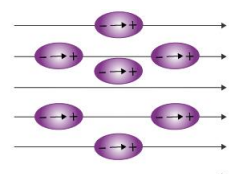


Dielectrics – The “Electric Sponges”

- Dielectrics contain bound charges that cannot move freely.
- When exposed to an electric field: Charges shift slightly → **polarization**.
- Induced dipoles produce an opposing field.
- Result: reduced net field inside.
- Used in capacitors, cables, and sensors.
- **Molecular Dipoles**
 - **Polar molecules:** permanent dipoles (e.g., water).
 - **Nonpolar molecules:** induced dipoles when field applied (e.g., oxygen, nitrogen).
 - **Dipole moment:** $p = qd$
 - Dipoles align to reduce net internal field.



Mechanisms of Polarization

Type	Mechanism	Example	Magnitude
Electronic 	Electron cloud shift	Argon	$10^{-15}m$
Ionic 	Ion displacement in solids	NaCl	$10^{-12}m$
Orientational 	Rotation of polar molecules	Water	$10^{-10}m$

The Polarization Vector (1/3)

- A polarized dielectric can be treated as combination of individual dipoles with dipole moments: $\vec{p} = Q\vec{d}$
- If there are N dipoles in a volume Δv of the dielectric, the total dipole moment due to the electric field is:

$$Q_1\vec{d}_1 + Q_2\vec{d}_2 + \dots + Q_N\vec{d}_N = \sum_{k=1}^N Q_k\vec{d}_k$$

- As a measure of intensity of the polarization, we define **polarization $\mathbf{P}$** ($\frac{C}{m^2}$) as the *dipole moment per unit volume* of the dielectric; that is:

$$\vec{P} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^N Q_k\vec{d}_k}{\Delta v}$$

- To compute the field due to a polarized dielectric, consider the dielectric material consisting of dipoles with dipole moment $\vec{P}$ per unit volume, the potential dV at an exterior point O due to the dipole moment $\vec{P}dv$ is

$$dV = \frac{\vec{P} \cdot \vec{a}_R dv'}{4\pi\epsilon_0 R^2}$$

where $R = |\vec{r} - \vec{r}'|$ and $\vec{a}_R = \vec{r} - \vec{r}'$. Exploiting the property $\nabla' \left(\frac{1}{R} \right) = \frac{\vec{a}_R}{R^2}$, we can rewrite $\frac{\vec{P} \cdot \vec{a}_R}{R^2} = \vec{P} \cdot \nabla' \left(\frac{1}{R} \right)$. Applying vector identity $\nabla' \cdot \left(\frac{\vec{P}}{R} \right) = \nabla' \cdot \left(\frac{\vec{P}}{R} \right) - \frac{\nabla' \cdot \vec{P}}{R}$. This implies:

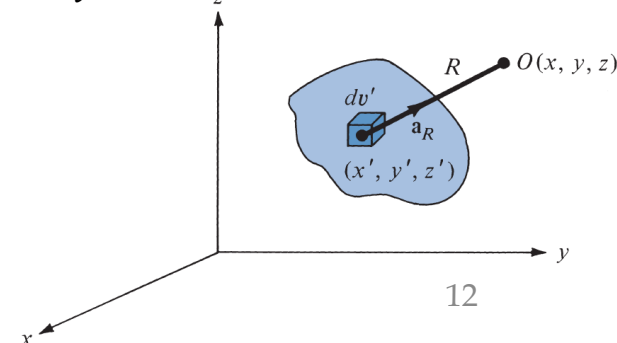
$$dV = \frac{1}{4\pi\epsilon_0} \left[\nabla' \cdot \left(\frac{\vec{P}}{R} \right) - \frac{\nabla' \cdot \vec{P}}{R} \right] dv'$$

- The potential due to all dipoles is then:

$$\begin{aligned} V &= \int_{v'} \frac{1}{4\pi\epsilon_0} \left[\nabla' \cdot \left(\frac{\vec{P}}{R} \right) - \frac{\nabla' \cdot \vec{P}}{R} \right] dv' \\ &= \frac{1}{4\pi\epsilon_0} \int_{v'} \nabla' \cdot \left(\frac{\vec{P}}{R} \right) dv' - \frac{1}{4\pi\epsilon_0} \int_{v'} \frac{\nabla' \cdot \vec{P}}{R} dv' \end{aligned}$$

Applying divergence theorem to the first part of the right hand side:

$$V = \frac{1}{4\pi\epsilon_0} \oint_{S'} \frac{\vec{P} \cdot \vec{a}'_n}{R} dS' + \frac{1}{4\pi\epsilon_0} \int_{v'} \frac{-\nabla' \cdot \vec{P}}{R} dv'$$



The Polarization Vector (2/3)

$$V = \frac{1}{4\pi\epsilon_0} \oint_{S'} \frac{\vec{P} \cdot \vec{a}'_n}{R} dS' + \frac{1}{4\pi\epsilon_0} \int_{v'} \frac{-\nabla' \cdot \vec{P}}{R} dv'$$

Recalling the potential due to surface and volume charge distributions $V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{S'} \frac{\rho_S}{|\vec{r}-\vec{r}'|} dS'$ and $V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{v'} \frac{\rho_v}{|\vec{r}-\vec{r}'|} dv'$ and comparing with equation above, we can establish the equivalence:

$$\rho_{ps} = \vec{P} \cdot \vec{a}'_n \text{ and } \rho_{pv} = -\nabla' \cdot \vec{P}$$

- We refer to ρ_{ps} and ρ_{pv} as *bound* (or *polarization*) *surface* and *volume charge densities*, respectively, as distinct from *free* surface and volume charge densities ρ_S and ρ_v .
- Bound charges are those that are not free to move within the dielectric material; they are caused by the displacement that occurs on a molecular scale during polarization.
- Free charges are those that are capable of moving over macroscopic distance, as do electrons in a conductor; they are the stuff we control.
- The total positive bound charge on surface S bounding the dielectric is:

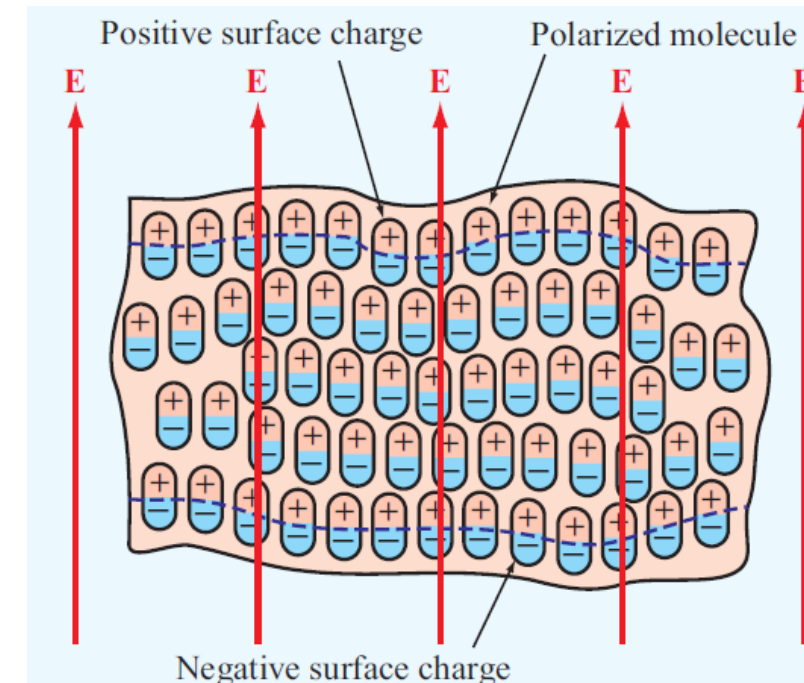
$$Q_b = \oint_S \vec{P} \cdot d\vec{S} = \oint_S \rho_{ps} dS$$

- while the charge that remains inside S is:

$$-Q_b = \int_v \rho_{pv} dv = - \int_v \nabla \cdot \vec{P} dv$$

- If the entire dielectric were electrically neutral prior to application of the electric field and if we have not added any free charge, the dielectric will remain electrically neutral. Thus the total charge of the dielectric material remains zero, that is:

$$\text{total charge} = \oint_S \rho_{ps} dS + \int_v \rho_{pv} dv = 0$$



The Polarization Vector (3/3)

- We now consider the case in which the dielectric region contains free charge. If ρ_v is the volume density of free charge, the total volume charge density ρ_t is given by:
$$\rho_t = \rho_v + \rho_{pv} = \nabla \cdot \epsilon_0 \vec{E}$$
Hence:
$$\begin{aligned}\rho_v &= \nabla \cdot \epsilon_0 \vec{E} - \rho_{pv} = \nabla \cdot \epsilon_0 \vec{E} + \nabla \cdot \vec{P} \\ &= \nabla \cdot (\epsilon_0 \vec{E} + \vec{P}) = \nabla \cdot \vec{D}\end{aligned}$$
where:
$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$
- We conclude that the net effect of the dielectric on the electric field $\vec{E}$ is to increase $\vec{D}$ inside it by the amount $\vec{P}$.
- In other words, the application of $\vec{E}$ to the dielectric material causes the flux density to be greater than it would be in free space.
- It should be noted that the definition of $\vec{D}$ in free space is a special case because $\vec{P} = 0$ in free space.



Dielectric Constant

- For some dielectrics, $\vec{P}$ is proportional to the applied electric field $\vec{E}$, and we have:

$$\vec{P} = \chi_e \varepsilon_0 \vec{E}$$

where χ_e , known as the **electric susceptibility** of the material, is more or less a measure of how susceptible (or sensitive) a given dielectric is to electric fields.

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \varepsilon_0 \vec{E} + \chi_e \varepsilon_0 \vec{E} = \varepsilon_0 (1 + \chi_e) \vec{E} = \varepsilon_0 \varepsilon_r \vec{E} = \varepsilon \vec{E}$$

where $\varepsilon = \varepsilon_0 \varepsilon_r$ and $\varepsilon_r = 1 + \chi_e = \frac{\varepsilon}{\varepsilon_0}$

- ε is called the **permittivity** of the dielectric, ε_0 is the permittivity of free space $\varepsilon_0 = 8.854 \times 10^{-12} \left[\frac{F}{m} \right]$, and ε_r is called the **dielectric constant** or **relative permittivity** of the medium.



Dielectric Strength

- The discussion dielectrics so far assumes ideal dielectrics.
- Practically speaking, no dielectric is ideal.
- When the electric field in a dielectric is sufficiently large, it begins to pull electrons completely out of the molecules, and the dielectric becomes conducting.
- **Dielectric breakdown** is said to have occurred when a dielectric becomes conducting.
- Dielectric breakdown occurs in all kinds of dielectric materials (gases, liquids, or solids) and depends on the nature of the material, temperature, humidity, and the amount of time that the field is applied.
- The minimum value of the electric field at which dielectric breakdown occurs is called the **dielectric strength** of the dielectric material.
 - Glass $\sim 35 \frac{MV}{m}$, Mica $\sim 70 \frac{MV}{m}$, Air $\sim 3 \frac{MV}{m}$
- Example: Lightning



Continuity Equation

- From the principle of charge conservation, the time rate of decrease of charge within a given volume must be equal to the net outward current flow through the surface of the volume.
- Thus, the current I_{out} coming out of the closed surface is:

$$I_{out} = \oint_S \vec{J} \cdot d\vec{S} = \frac{-dQ_{in}}{dt}$$

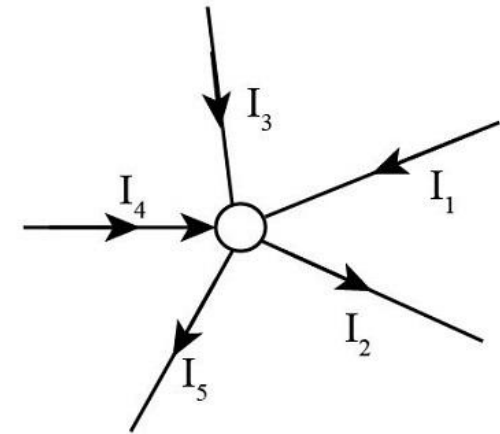
where Q_{in} is the total charge enclosed by the closed surface. Using the divergence theorem, we can write:

$$\oint_S \vec{J} \cdot d\vec{S} = \int_v \nabla \cdot \vec{J} dv, \text{ hence } \frac{-dQ_{in}}{dt} = -\frac{d}{dt} \int_v \rho_v dv = -\int_v \frac{\partial \rho_v}{\partial t} dv$$

- Therefore:

$$\int_v \nabla \cdot \vec{J} dv = -\int_v \frac{\partial \rho_v}{\partial t} dv \Rightarrow \nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}$$

- This is called the **continuity of current equation** or just **continuity equation**.
- This equation is derived from the principle of conservation of charge and essentially states that there can be no accumulation of charge at any point.
- For steady currents, $\frac{\partial \rho_v}{\partial t} = 0$, and hence $\nabla \cdot \vec{J} = 0$, showing that the total charge leaving a volume is the same as the total charge entering it.
- Kirchhoff's current law** follows from this.



Boundary Conditions: The Rules of Interface

- If a field exists in a region consisting of two different media, the conditions that the field must satisfy at the interface separating the medium are called boundary conditions.
- The conditions will be dictated by the types of material the media are made of. We will analyze an interface separating:
 - Dielectric 1 and dielectric 2
 - Conductor and dielectric
 - Conductor and free space

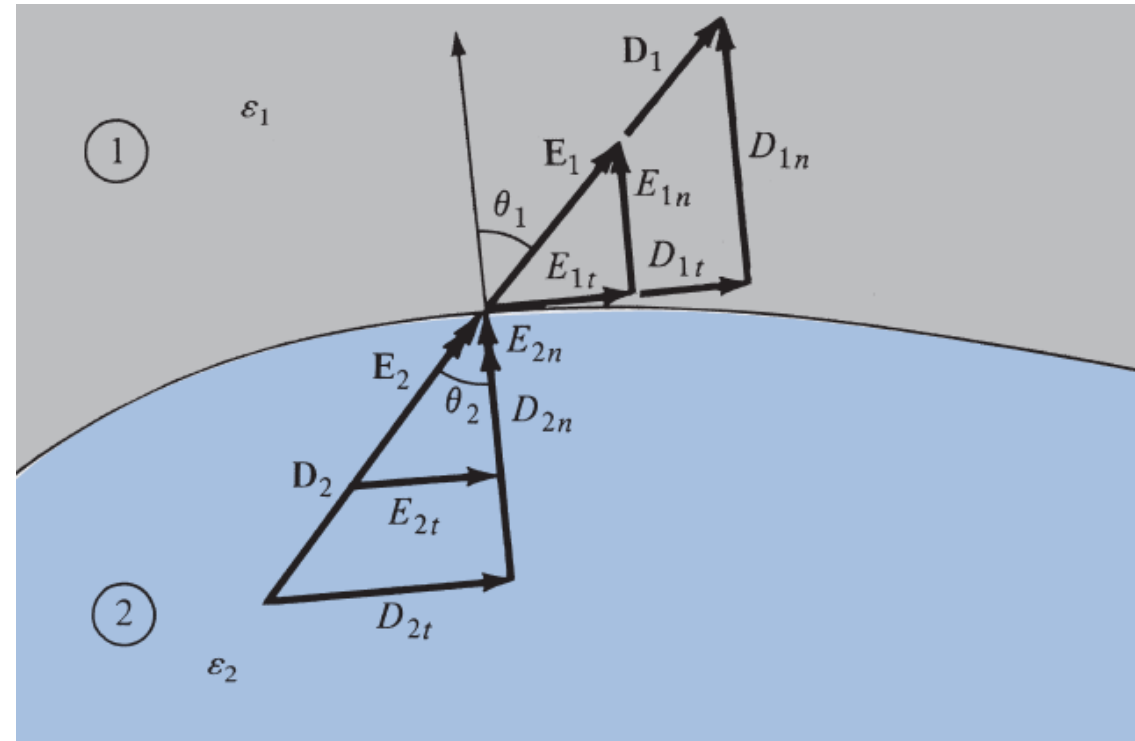
- To determine the boundary conditions, we will use Maxwell's equations:

$$\oint_L \vec{E} \cdot d\vec{l} = 0 \text{ and } \oint_S \vec{D} \cdot d\vec{S} = Q_{enc}$$

- where Q_{enc} is the free charge enclosed by the surface S . Also we need to decompose the electric field intensity $\vec{E}$ into two orthogonal components:

$$\vec{E} = \vec{E}_t + \vec{E}_n$$

- where $\vec{E}_t$ and $\vec{E}_n$ are, respectively, the tangential and normal components of $\vec{E}$ to the interface of interest.



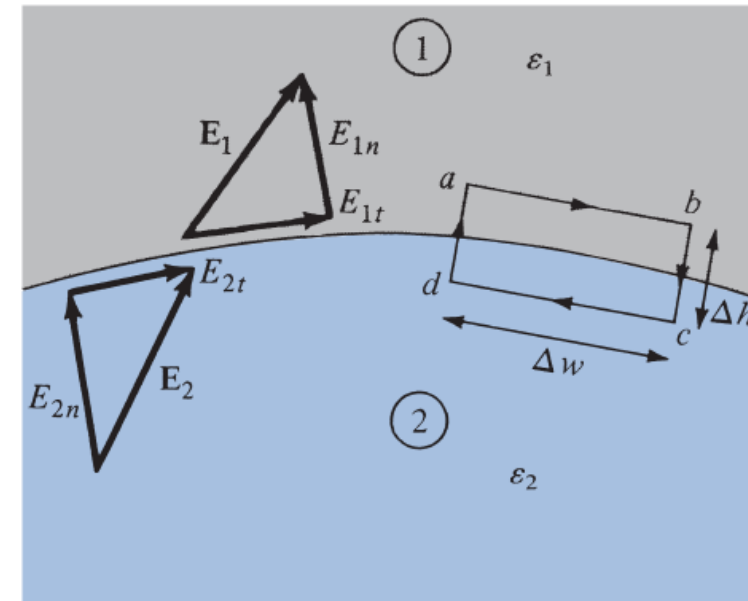
Dielectric-Dielectric Boundary Conditions (1/3)

- Consider the $\vec{E}$ field existing in a region that consists of two different dielectrics characterized by $\epsilon_1 = \epsilon_0 \epsilon_{r1}$ and $\epsilon_2 = \epsilon_0 \epsilon_{r2}$
- The fields $\vec{E}_1$ and $\vec{E}_2$ can be decomposed as: $\vec{E}_1 = \vec{E}_{1t} + \vec{E}_{1n}$ and $\vec{E}_2 = \vec{E}_{2t} + \vec{E}_{2n}$
- If we apply $\oint_L \vec{E} \cdot d\vec{l} = 0$ to the closed path $abcd$, assuming that the path is very small with respect to the spatial variation of $\vec{E}$, we obtain:

$$0 = E_{1t}\Delta w - E_{1n}\frac{\Delta h}{2} - E_{2n}\frac{\Delta h}{2} - E_{2t}\Delta w + E_{1n}\frac{\Delta h}{2} + E_{2n}\frac{\Delta h}{2} \Rightarrow 0 = (E_{1t} - E_{2t})\Delta w$$

$$E_{1t} = E_{2t} \text{ and } \frac{D_{1t}}{\epsilon_1} = \frac{D_{2t}}{\epsilon_2}$$

- The tangential components of $\vec{E}$ are the same on the two sides of the boundary.
- In other words, $\vec{E}_t$ undergoes no change on the boundary and it is said to be **continuous** across the boundary.
- While $\vec{D}_t$ is said to be **discontinuous** across the interface.



Dielectric-Dielectric Boundary Conditions (2/3)

- Let us now apply Gauss's law to the cylindrical Gaussian surface below.
- The contribution due to the sides vanishes.
- When $\Delta h \rightarrow 0$,

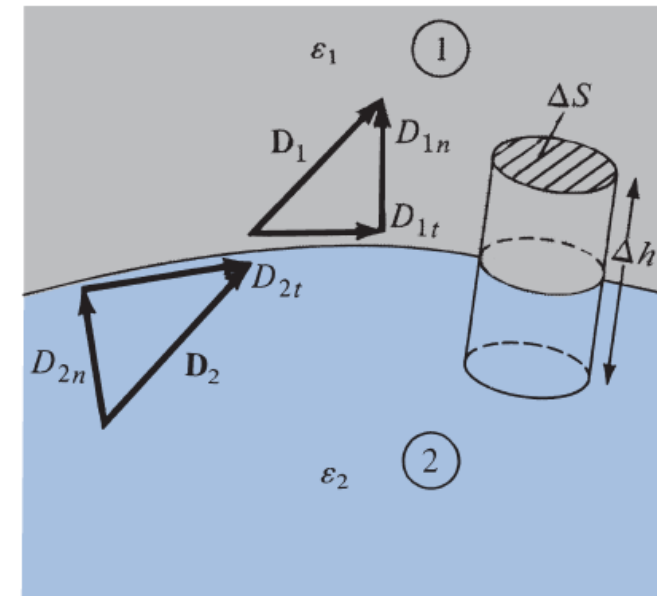
$$\Delta Q = \rho_S \Delta S = D_{1n} \Delta S - D_{2n} \Delta S$$

$$D_{1n} - D_{2n} = \rho_S$$

- where ρ_S is the free charge density placed deliberately at the boundary.
- If no free charges exist at the interface (i.e., charges are not deliberately placed there which is typical of dielectric-dielectric boundaries), $\rho_S = 0$ and we have:

$$D_{1n} = D_{2n} \text{ and } \epsilon_1 E_{1n} = \epsilon_1 E_{2n}$$

- The normal component of $\vec{D}$ is **continuous** across the interface while the normal component of $\vec{E}$ is **discontinuous**.



Dielectric-Dielectric Boundary Conditions (3/3)

- We can use the boundary conditions to determine the “refraction” of the electric field across the interface.
- Consider $\vec{E}_1$ and $\vec{E}_2$ making angles θ_1 and θ_2 with the *normal* to the interface. Since the tangential component of the field is continuous, we have

$$E_1 \sin \theta_1 = E_{1t} = E_{2t} = E_2 \sin \theta_2 \Rightarrow E_1 \sin \theta_1 = E_2 \sin \theta_2$$

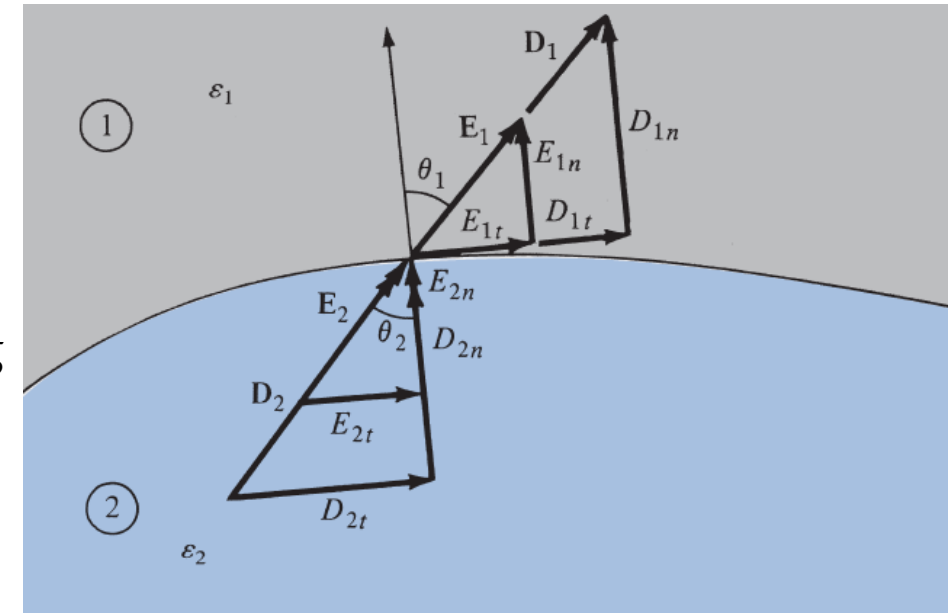
- Considering the normal component of the electric field for a charge free interface:

$$\epsilon_1 E_1 \cos \theta_1 = D_{1n} = D_{2n} = \epsilon_2 E_2 \cos \theta_2 \Rightarrow \epsilon_1 E_1 \cos \theta_1 = \epsilon_2 E_2 \cos \theta_2$$

- Taking the ratio of the two equations:

$$\frac{E_1 \sin \theta_1}{\epsilon_1 E_1 \cos \theta_1} = \frac{E_2 \sin \theta_2}{\epsilon_2 E_2 \cos \theta_2} \Rightarrow \frac{\tan \theta_1}{\epsilon_1} = \frac{\tan \theta_2}{\epsilon_2}$$
$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}$$

- This is the **law of refraction** of the electric field at a boundary free of charge.
- In general, an interface between two dielectrics produces bending of the flux lines as a result of unequal polarization charges that accumulate on the opposite sides of the interface.



Conductor–Dielectric Boundary Conditions

- The conductor is assumed to be perfect (i.e., $\sigma \rightarrow \infty$ and $\rho_c = 0$). Inside the conductor, the electric field vanishes. $\vec{E}_c = \vec{E}_{ct} + \vec{E}_{cn} = 0$. Applying $\oint_L \vec{E} \cdot d\vec{l} = 0$ to the closed path $abcd$:

$$0 = 0 \cdot \Delta w - 0 \cdot \frac{\Delta h}{2} - E_n \frac{\Delta h}{2} - E_t \Delta w + 0 \cdot \frac{\Delta h}{2} + E_n \frac{\Delta h}{2}$$

$$E_t = 0$$

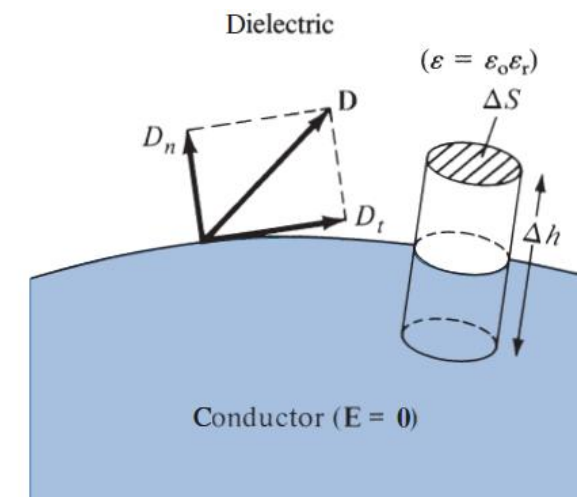
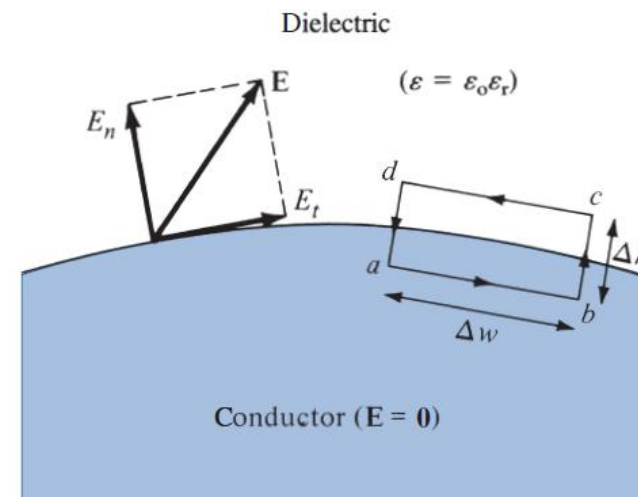
- Similarly, applying Gauss's law to the cylindrical Gaussian surface:

$$\Delta Q = \rho_s \Delta S = D_n \Delta S - 0 \cdot \Delta S$$

$$D_n = \rho_s$$

- Thus under static conditions, the following conclusions can be made about a perfect conductor:

- No electric field may exist *within* a conductor
- Since $\vec{E} = -\nabla V = 0$, there can be no potential difference between any two points in the conductor; that is, a conductor is an equipotential body.
- An electric field $\vec{E}$ must be external to the conductor and must be **normal** to its surface



Classroom exercises

Scenario: A parallel-plate capacitor contains polystyrene, which has a relative permittivity of $\epsilon_r = 2.55$. The uniform electric field intensity inside the dielectric is $\mathbf{E} = 10 \text{ kV/m}$. The plate separation is 1.5 mm.

Tasks:

- Calculate the **Electric Flux Density ($\mathbf{D}$)**.
- Calculate the **Polarization ($\mathbf{P}$)** vector magnitude.
- Determine the **free surface charge density (ρ_s)** on the plates.
- Find the **potential difference (V)** between the plates.



Solution to Exercise 1

Given: $\epsilon_r = 2.55$, $E = 10 \text{ kV/m} = 10^4 \text{ V/m}$, $d = 1.5 \text{ mm} = 1.5 \times 10^{-3} \text{ m}$.

Constant: $\epsilon_0 \approx 8.854 \times 10^{-12} \text{ F/m}$.

a. Electric Flux Density (D)

The relationship between D and E in a linear dielectric is $D = \epsilon_0 \epsilon_r E$.

$$D = (8.854 \times 10^{-12} \text{ F/m}) \times 2.55 \times (10^4 \text{ V/m})$$

$$D \approx \mathbf{2.257 \times 10^{-7} \text{ C/m}^2}$$

b. Polarization (P)

The polarization vector relates D and E by $D = \epsilon_0 E + P$.

$$\begin{aligned} P &= D - \epsilon_0 E = \epsilon_0 (\epsilon_r - 1) E \\ P &= (8.854 \times 10^{-12}) \times (2.55 - 1) \times (10^4) \\ P &\approx \mathbf{1.372 \times 10^{-7} \text{ C/m}^2} \end{aligned}$$

c. Free Surface Charge Density (ρ_s)

The normal component of the Electric Flux Density (D_n) at the surface of the conductor (plate) equals the free surface charge density (ρ_s). Since the field is normal to the plates, $D_n = D$.

$$\rho_s = D \approx \mathbf{2.257 \times 10^{-7} \text{ C/m}^2}$$

d. Potential Difference (V)

For a uniform electric field, the potential difference is the product of the field strength and the separation distance.

$$\begin{aligned} V &= Ed \\ V &= (10^4 \text{ V/m}) \times (1.5 \times 10^{-3} \text{ m}) \\ V &= \mathbf{15 \text{ V}} \end{aligned}$$



Classroom Exercise 2: Field Refraction at a Boundary

Scenario: Two charge-free, homogeneous dielectrics meet at the $z = 0$ plane.

- Medium 1 ($z \geq 0$): $\epsilon_{r1} = 4$
- Medium 2 ($z < 0$): $\epsilon_{r2} = 3$

The electric field in Medium 1 is given by $E_1 = 1\hat{a}_x - 2\hat{a}_y + 3\hat{a}_z$ kV/m.

Tasks:

- Find the **Electric Field** E_2 in Medium 2.
- Verify the **Law of Refraction** for the angles θ_1 and θ_2 the fields make with the normal direction ($\hat{a}_z$):

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}$$



Solution Exercise 2 (1/2)

Given: $\epsilon_{r1} = 4$, $\epsilon_{r2} = 3$, $\rho_s = 0$. $E_1 = 1\hat{a}_x - 2\hat{a}_y + 3\hat{a}_z$ kV/m.

- The interface is the $z = 0$ plane. The **normal vector** ($\hat{n}$) is $\hat{a}_z$.
- The **tangential components** (E_t) are those in the $\hat{a}_x$ and $\hat{a}_y$ directions.
- The **normal component** (E_n) is in the $\hat{a}_z$ direction.

a. Find E_2

1. **Tangential E is Continuous:**

$$E_{2t} = E_{1t}$$

$$E_{2x} = E_{1x} = 1 \text{ kV/m}$$

$$E_{2y} = E_{1y} = -2 \text{ kV/m}$$

$$E_{2t} = 1\hat{a}_x - 2\hat{a}_y \text{ kV/m}$$

2. **Normal D is Continuous ($\rho_s = 0$):**

$$D_{1n} = D_{2n} \implies \epsilon_0 \epsilon_{r1} E_{1n} = \epsilon_0 \epsilon_{r2} E_{2n}$$

$$\epsilon_{r1} E_{1n} = \epsilon_{r2} E_{2n}$$

3. **Calculate E_{2n} :**

$$E_{2n} = \frac{\epsilon_{r1}}{\epsilon_{r2}} E_{1n} = \frac{4}{3} \times (3 \text{ kV/m})$$

$$E_{2n} = 4 \text{ kV/m}$$

4. **Assemble E_2 :**

$$E_2 = E_{2t} + E_{2n}\hat{a}_z$$

$$E_2 = (1\hat{a}_x - 2\hat{a}_y + 4\hat{a}_z) \text{ kV/m}$$



Solution Exercise 2 (2/2)

b. Verify the Law of Refraction

The angle θ an electric field makes with the normal is given by $\tan \theta = \frac{|E_t|}{E_n}$.

1. Calculate $\tan \theta_1$ and $\tan \theta_2$:

$$\tan \theta_1 = \frac{|E_{1t}|}{E_{1n}} = \frac{\sqrt{1^2 + (-2)^2}}{3} = \frac{\sqrt{5}}{3}$$

$$\tan \theta_2 = \frac{|E_{2t}|}{E_{2n}} = \frac{\sqrt{1^2 + (-2)^2}}{4} = \frac{\sqrt{5}}{4}$$

2. Calculate the Ratio:

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\sqrt{5}/3}{\sqrt{5}/4} = \frac{4}{3}$$

3. Compare with Permittivity Ratio:

$$\frac{\epsilon_{r1}}{\epsilon_{r2}} = \frac{4}{3}$$

The Law of Refraction is **verified**.



Summary and Concept Map

- Materials influence electric fields via **currents** and **polarization**.
 - $\vec{P}, \vec{E}, \vec{D}, \epsilon$ are interlinked:

$$\vec{D} = \epsilon \vec{E} = \epsilon_0 \vec{E} + \vec{P}$$

- Bound charges alter net fields.
- Boundary conditions ensure continuity.
- Tangential component of electric field is continuous.
- Normal component electric flux density changes by an amount equal to surface charge density at the boundary.
- An electric field $\vec{E}$ must be external to the conductor and must be **normal** to its surface.



End of slide

